

Predicting Student Success Using Different Regression Analyses

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Motivation and Objective

- Interested in student academic success and wanting to help, but not sure how
- Want to understand the factors that may affect a student's academic success
- Factors include environmental factors, motivation, family background and more
- Our objective is to analyze the most influential factors using the chosen dataset, and to then build a model predicting a student's success given data describing those factors relating to them



be helpful for faculty in universities trying to gauge whether a student will fail or succeed in an exam, and ways in which they could be



Our Dataset

- Open student dataset of size 1000 from Kaggle
- Contains information about their academic history, study habits, lifestyle factors, psychological factors and family background
- Factors include environmental factors, motivation, family background and more



Methodology – Part 1

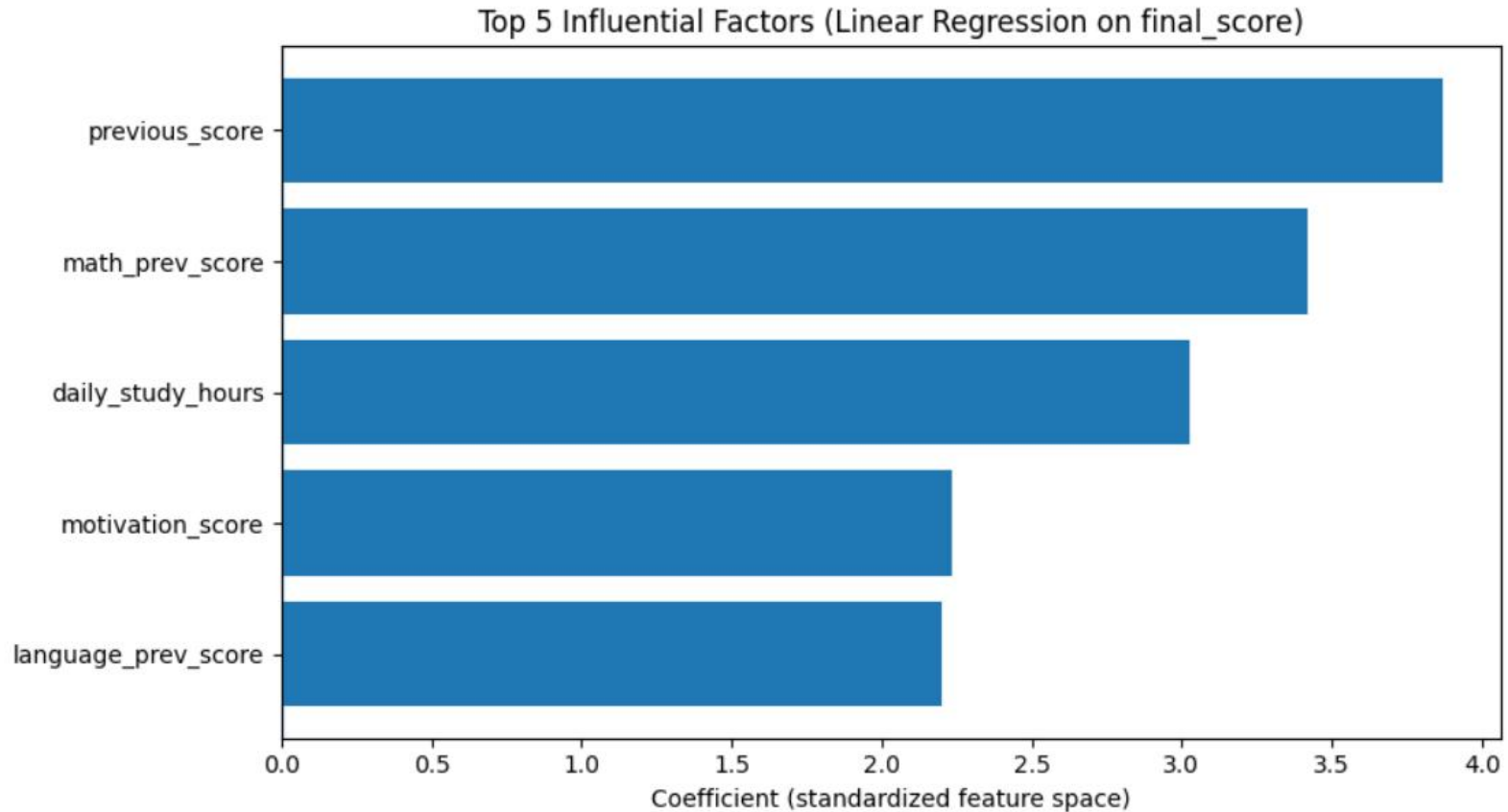
- Used different statistical analysis methods to analyze what the dataset is telling us
- Used linear, lasso, and logistic regression models to show predicted performance based on the given factors
- Ran the model while having removed the prior performance factor, and ran another with just actionable factors such as motivation and study habits
- Generated visual plots for the different regression models to showcase the relations between the different factors
- Evaluated each regression model through statistical methods to ensure their efficiency and accuracy

Methodology – Part 2

- Implemented four different kinds of machine learning packages (KNN, Random Forest, Gradient Boosting, XGBoost) to assess a student's academic performance, using the dataset
- Utilized features deemed as actionable features from part 1 to implement ML models
- Created an identifier trying to find which is the best method for predicting a student's performance outside of the dataset
- Evaluated and compared the ML packages using classification metrics (Accuracy, F1-Score, AUC-ROC)
- Used SHAP model to make sense of results from our chosen ML model, to give advice to individual students based on the output factor significances

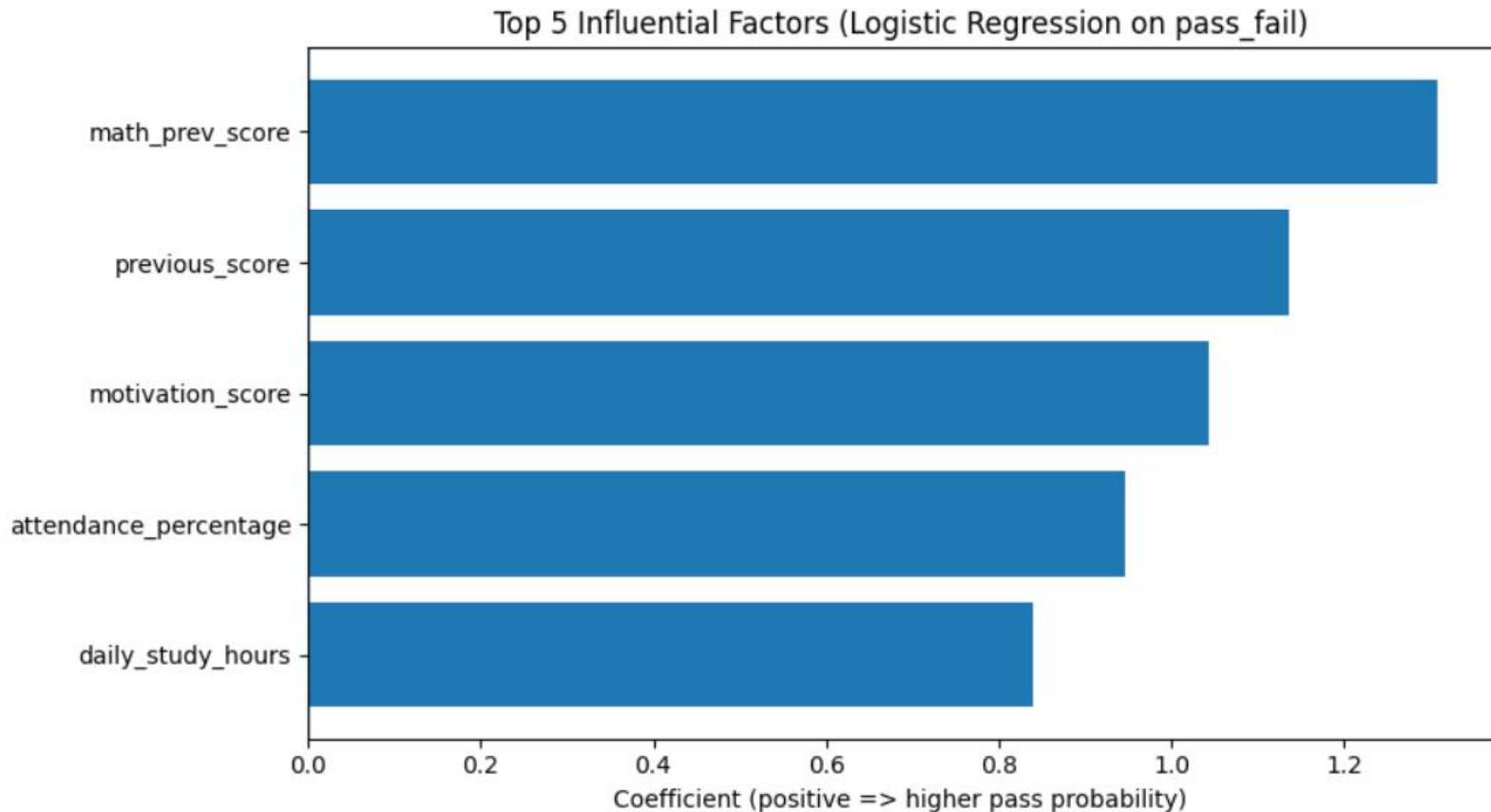
Our Results

Linear Regression Model



- Based on our linear regression model, previous scores do tend to be the most influential factors, as well as time spent studying

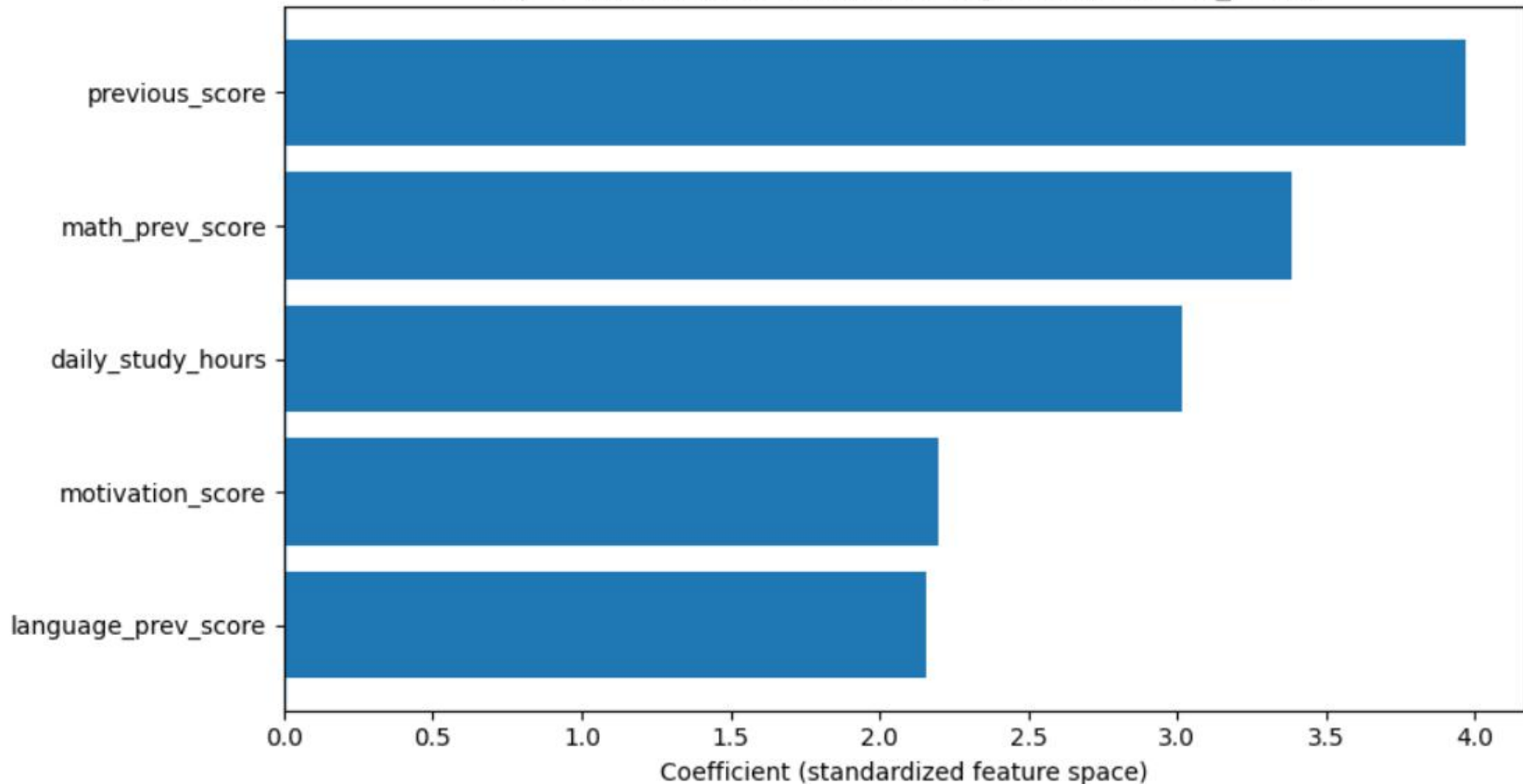
Logistic Model



- The logistic model shows us similar results, putting previous scores at the top in terms of correlation to passing. It still highlights a high positive correlation between the actionable factors, however it magnifies the correlation between the past scores and future ones.

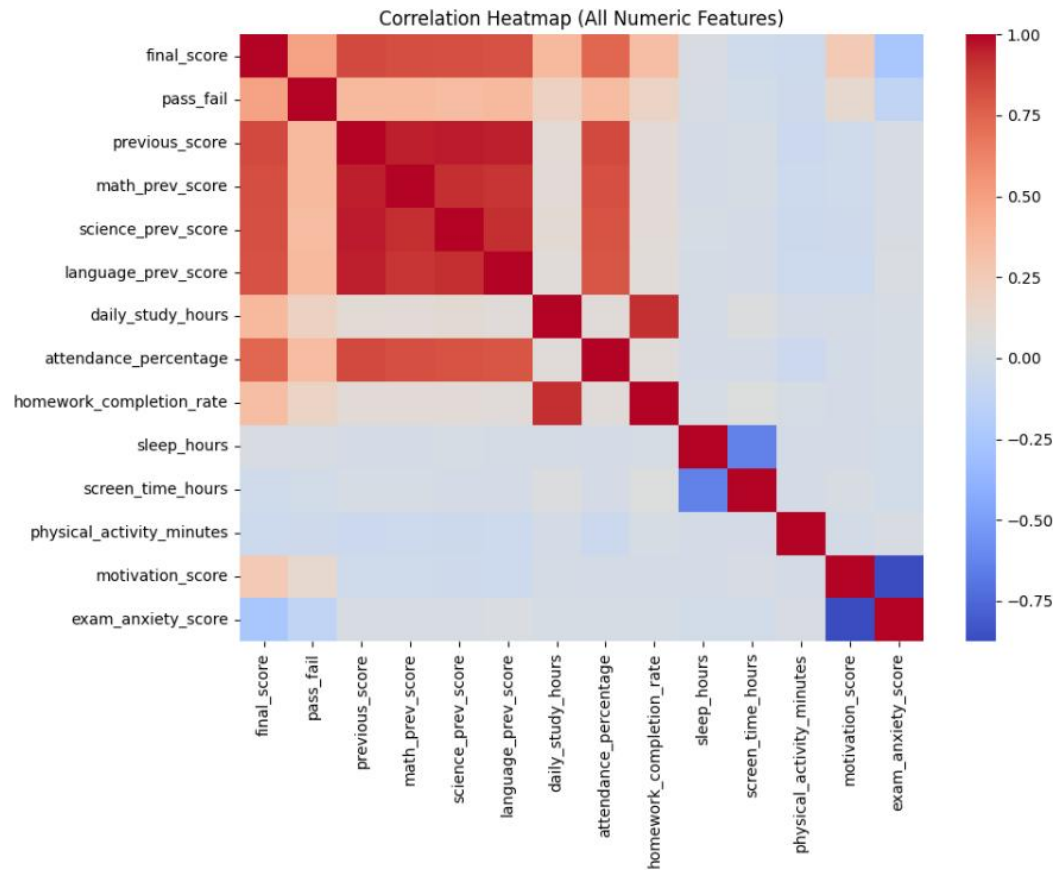
Lasso Model

Top 5 Influential Factors (Lasso Regression on final_score)



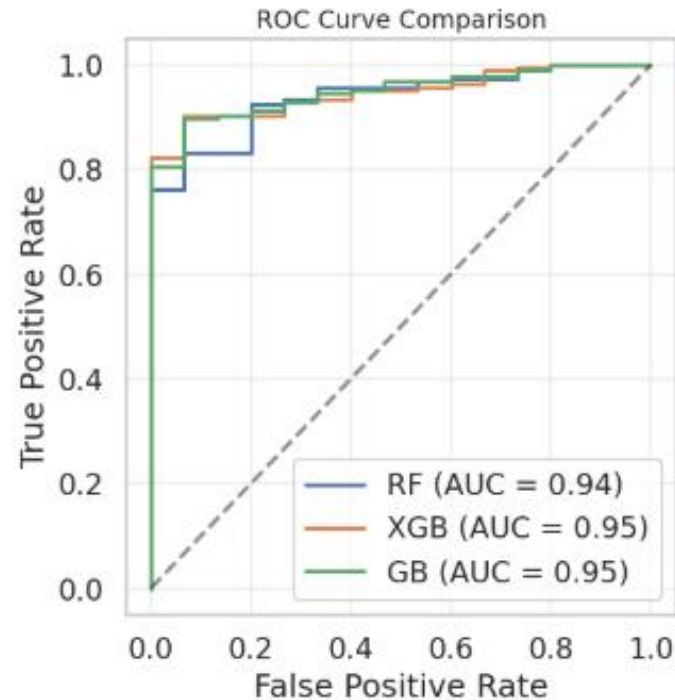
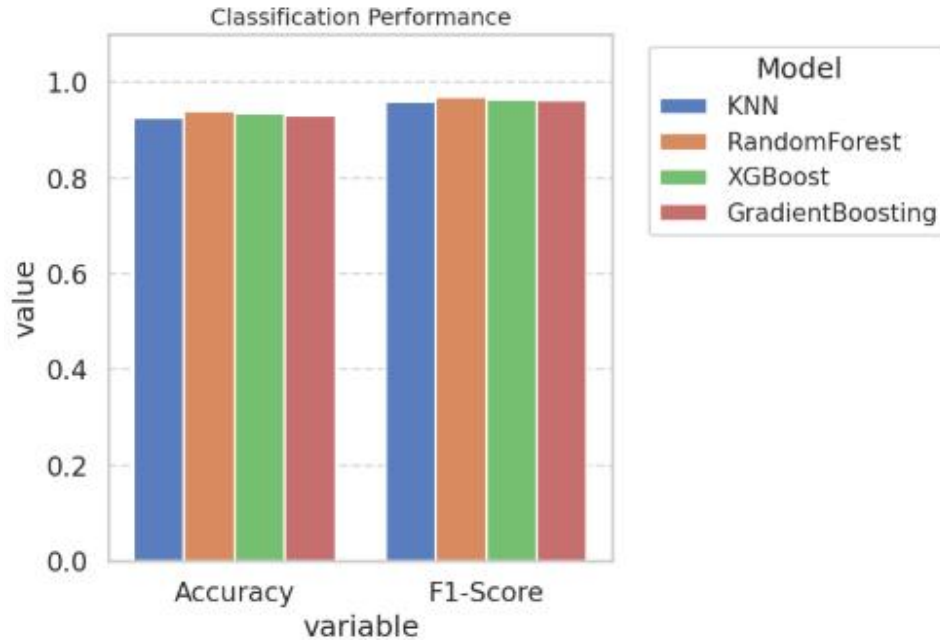
- The Lasso model is a stricter model, showing the importance of each factor. We can see that, once again, the previous scores are highlighted as the most important for prediction.

Heat Map



- This heatmap shows the correlation between the different factors, with the redder colors showing a higher positive correlation. We can see that the highest correlation is between the previous scores and the final score, further supporting the trend that previous scores are the largest indicators of the student's future score.

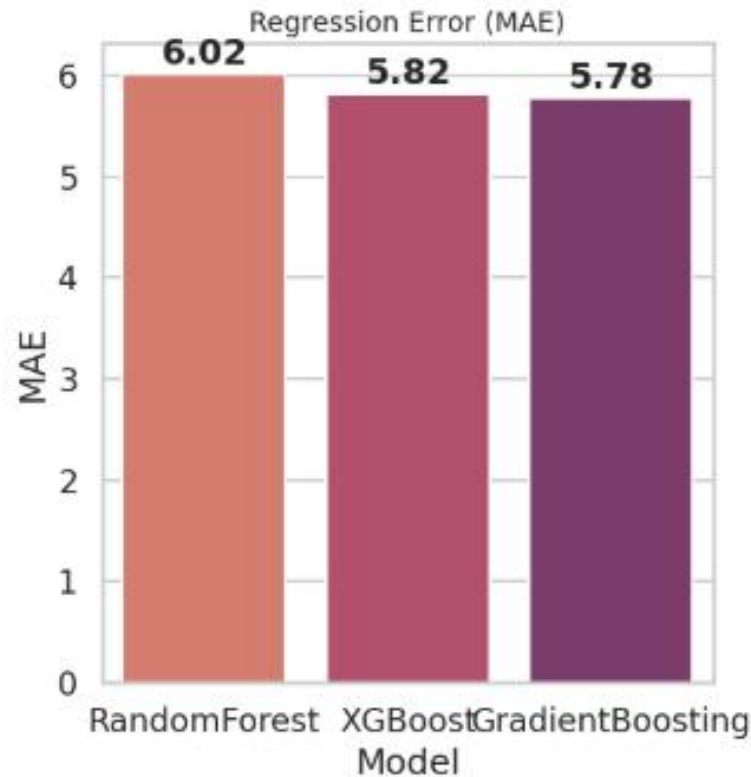
Performance of Model Packages



- These plots show the performance of each ML package, categorized by their accuracy, F1-score and AUC-ROC
- Although RandomForest had the highest classification accuracy, XGBoost performed better in the AUC-ROC and comparable in F1-Score, and had a lower regression error.

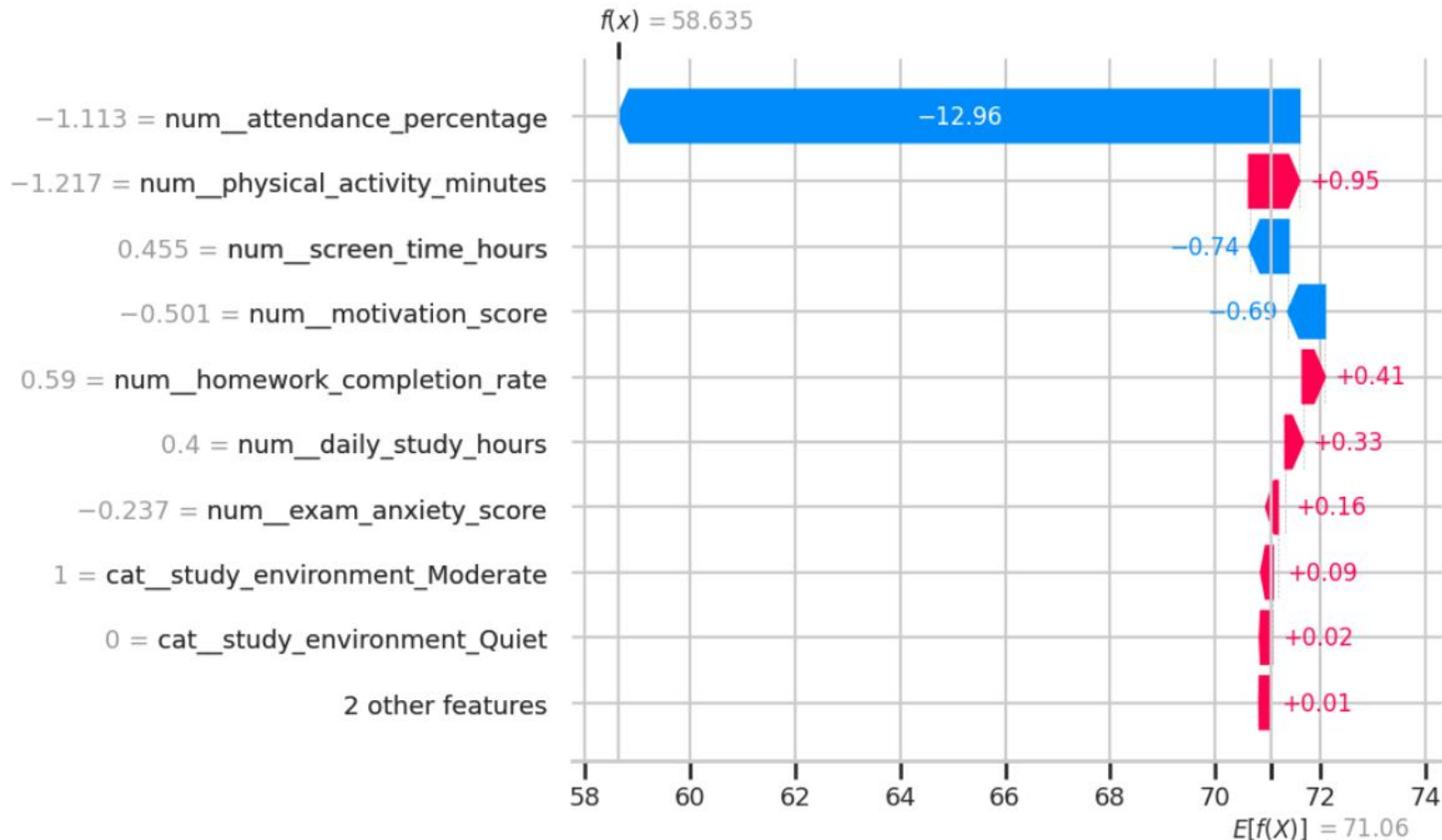
	Model	Accuracy	F1-Score	AUC-ROC
0	KNN	0.925	0.961039	0.500000
1	RandomForest	0.940	0.968421	0.937658
2	XGBoost	0.935	0.965517	0.946667
3	GradientBoosting	0.930	0.962567	0.948468

Regression Error



- This graph shows the calculated regression errors of each ML package. Lower is better, and we can see how the XGBoost model has one of the lowest errors, contribution to our decision in choosing the XGBoost as our go to model.

Output from SHAP model



- This SHAP model evaluates the outputs from the chosen XGBoost model for one student. As seen from above, these are the significances of each factor influencing the student's score. The attendance of the student was the most influential factor, in this case having a huge negative impact on their future predicted score.

**THANK
YOU**